MIL-C-43232A

19 May 1971

SUPERSEDING
MIL-C-43232
3 June 1964

MILITARY SPECIFICATION

COMPOSING MACHINES, PHOTO-PRINTING TYPE: SINGLE LINE AND MULTI-LINE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers three types and two classes of general purpose, photo-printing type composing machines (see 6.1).

1.2 Classification.- Composing machines shall be of the following types and classes as specified (see 6.2 and 6.8):

Type I	-	Composing and processing machine producing single line copy on 35 or 51 millimeter photosensitized paper and film.
Type II	-	Composing and processing machine producing multiple line copy on 35 or 51 millimeter photosensitized paper and film.
Type III	-	Composing machine with separate processing unit producing line copy on 8 inch photosensitized paper and film.
Class A	-	Field use.
Class B	-	Office use.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 7430

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SPECIFICATIONS

FEDERAL

CC-M-636	-	Motors, Fractional Horsepower (Alternating Current).
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon.
TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple Wall.

MILITARY

MIL-P-116	-	Preservation, Methods Of.
MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Material.
MIL-L-10547	-	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Water-Proof, Flexible.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristic, Requirements For Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristic, Measurement Of.

DRAWING

NATICK LABORATORIES

6-1-5509	-	Composing Machines, Photo Printing Type: Single Line and Multiple Line; Symbols.

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(Miniature copy of Drawing 6-1-5509, identified as figure 1, is attached for information purposes only).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Applicable UL Standards

(Application for copies of applicable UL standards should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, Ill. 60611; 333 Pfingsten Rd., Northbrook, Ill. 60062; 1285 Walt Whitman Rd., Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specification and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2) the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.3).

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3.2 Standard product.- The composing machines covered by this specification shall be the manufacturers' standard commercial product, except for any changes necessary to conform to this specification.

3.3 Standards compliance.-

3.3.1 Safety standard.- The electrical components shall conform to the requirements of applicable Underwriters' Laboratories, Inc., standards.

* 3.3.2 Certification.- Prior to the approval of the first article or if none is submitted, prior to the approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer that the electrical components he proposes to supply under this specification meet the requirements of applicable UL standards. Acceptable evidence of meeting UL requirements shall be the UL label or UL listing mark on the component or a certified test report from a nationally recognized independent testing laboratory, approved by the contracting officer, stating that the electrical components offered have been tested and conform to applicable UL standards.

* 3.4 Materials.- Materials shall be as specified herein. Materials not definitely specified shall be the same as normally used by the manufacturer in his standard product provided the completed item complies with all provisions of this specification. Materials shall be either corrosion-resistant or treated to resist corrosion.

3.4.1 Plastic.- The plastic parts shall be inert to all photographic chemicals and, when applicable, shall be capable of withstanding the temperature and humidity storage tests specified in 4.4.1, and the rigidity and transportability test specified in 4.4.2.

3.4.2 Steel, sheet, low carbon.- Low carbon steel sheet shall be cold rolled, commercial quality, with a chemical composition between 1009 and 1020, of QQ-S-698.

* 3.4.3 Enamel.- Enamel shall be gray color No. 26134 of TT-E-529 and FED-STD-595.

* 3.5 Performance characteristics.- All machines shall be capable of photocomposition by exposing photosensitized paper and film to a negative film or disc type lettering font varying in type size from 10 point to 72 point. The copy produced shall be clean and uniform with defined edges free from fogging and light spreading when tested as specified in 4.4.3.1 and 4.4.3.2. There shall be no evidence of controls binding or other malfunction when tested as specified in 4.4.6.

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* 3.5.1 Class A machines.- Class A machines shall be capable of passing the temperature and humidity storage test specified in 4.4.1, and the rigidity and transportability tests specified in 4.4.2. The Government shall conduct the test specified in 4.4.2 (see 6.5).

* 3.6 Design.- The class A machines shall be designed to operate under field conditions (will withstand the requirements specified in 3.5.1), shall be equipped for electromagnetic compatibility (see 3.7.1.3), and shall have the electrical circuitry protected from fungus and moisture (see 3.7.1.4). The class B machines shall be designed for office use.

3.6.1 Type I.- The type I machine shall be a table model with integral processing units. The machine shall be capable of producing single line copy on 35 or 51 millimeter photosensitized paper and film as specified (see 6.2), when tested as specified in 4.4.3.1 and 4.4.3.2.

3.6.2 Type II.- The type II machines shall be either table or floor model composing machines as specified (see 6.2). The processing units shall be either an integral or an external type, as specified (see 6.2). The machines shall be capable of producing multiple line copy on 35 or 51 millimeter photosensitized paper and film, as specified (see 6.2), when tested as specified in 4.4.3.1 and 4.4.3.2.

* 3.6.3 Type III.- The type III machines shall be table model composing machines with separate processing units. The machines shall be capable of producing line copy on an 8 inch wide roll of photosensitized paper and film when tested as specified in 4.4.3.1 and 4.4.3.2.

3.6.4 Details of components.- All components subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machines shall be designed to incorporate the applicable features, controls, and devices specified hereinafter.

3.6.4.1 Letter and word spacing.- The machines shall automatically space letters and words, once the spacing has been manually preset. The machine shall be capable of adjusting the space between letters and the space between the words when tested as specified in 4.4.3.1 and 4.4.3.2.

3.6.4.2 Light source.- The light source for exposing the photosensitized paper and film shall be located so that the type lettering fonts can be changed without exposing the paper and film when tested as specified in 4.4.3.1 and 4.4.3.2.

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3.6.4.2.1 Exposure light and time control.- The machines shall be provided with a means to adjust either the intensity of the exposure light, or the length of time the exposure light remains lit when tested as specified in 4.4.3.1 and 4.4.3.2.

3.6.4.2.2 Exposure meter.- When specified (see 6.2), the machine shall be provided with an electronic, direct reading meter to indicate the intensity of the exposure light, and an adjusting rheostat to control the intensity of the exposure light when tested as specified in 4.4.4.

3.6.4.3 Paper and film cutting device.- The type I and type II machines shall be provided with a built-in, manually operated, paper and film cutting device (see 4.4.3.1 and 4.4.3.2). The type III machine shall be provided with a metal cutting edge that allows the paper or film to be manually torn off.

* 3.6.4.4 Foreign language composition.- The machines shall be designed to permit diacritical marks to be printed over characters in foreign language composition when tested as specified in 4.4.3.1 and 4.4.3.2 (see 6.6).

3.6.4.5 Photosensitized paper and film feed.-

3.6.4.5.1 Type I and type II machines.- The type I and type II machines shall be designed to automatically feed both photosensitized paper and film. In addition, both types of machines shall be provided with a control to allow manual feed of photosensitized paper and film when tested as specified in 4.4.3.1 and 4.4.3.2. When changing the paper or film roll, not more than 18 inches of the new roll shall be exposed as a result of installation in the machines when tested as specified in 4.4.3.2.

3.6.4.5.2 Type III machine.- The type III machine shall be designed to manually feed both photosensitized paper and film.

* 3.6.4.5.3 Sensitized paper or film capacity.- The machines shall be designed to use photosensitized paper and film either with a plain back or adhesive back of the following sizes:

Type I and type II machines:

Paper roll length, plain	- 100 feet
Paper roll length, adhesive back	- 75 feet
Film roll length, plain	- 50 feet
Film roll length, adhesive back	- 35 feet
Paper and film roll width	- 35 or 51 millimeter (see 3.6.1 and 3.6.2)
Paper thickness	- 0.006 - 0.010 inch
Film thickness	- 0.005 - 0.007 inch

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Type III machine:

Paper roll length	- 25 feet
Film roll length	- 15 feet
Paper and film roll width	- 8 inches
Paper thickness	- 0.006 - 0.010 inch
Film thickness	- 0.005 - 0.007 inch

3.6.4.6 Photosensitized paper and film processing.-

3.6.4.6.1 Type I machine.- The type I machine shall automatically process the exposed paper and film through a processing unit designed to contain photographic type-setting chemicals when tested as specified in 4.4.3.1 and 4.4.3.2. The processing unit shall be constructed of the plastic material specified in 3.4.1.

3.6.4.6.2 Type II machine.- The type II composing machine shall process the exposed paper and film either through an automatic processing tank or shall have separate manual developing facilities (see 6.2) designed to contain developer, fix, washwater or monobath solutions as specified. The automatic processing unit shall include a removable, compartmented, plastic tank to contain solutions required for developing and fixing exposed paper or film. The machine and tank shall be an integral unit designed to automatically process the exposed film or paper and to deliver slightly damp, finish copy outside the machine when tested as specified in 4.4.3.1 and 4.4.3.2. The processing unit shall be constructed of the plastic material specified in 3.4.1.

3.6.4.6.3 Type III machine.- A separate processing unit shall be furnished with the type III composing machine. The unit shall be motor driven, have a two compartment plastic tank, be designed to automatically develop and fix exposed film or paper up to 8 inches wide, deliver slightly damp copy within 2 minutes after it has been inserted into the unit, and shall produce copy free from fogging or light spreading when tested as specified in 4.4.3.3.

3.6.4.7 Indicator devices.-

3.6.4.7.1 Type I and type II machines.- The type I and type II machines shall be equipped with an indicator light to indicate either "exposure" or "developing in process." The "exposure" light shall become lit when the light source is energized. The "developing in process" light shall become lit when the exposed paper or film enters the processsing unit.

3.6.4.7.2 Type III machine.- The type III machine shall be equipped with two warning lights; one to indicate "machine on" and one to indicate "paper out." In addition, there shall be a sound device to warn the operator when the machine is nearing the end of a line of composition.

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3.6.4.8 Accessibility of interior components.- All major interior components and installed attachments shall be accessible for maintenance, repair and replacement without removal of other major assemblies and sub-assemblies.

3.6.4.8.1 Class A machines.- Covers and access panels that require removal for adjustment of interior components shall be equipped with hand-operable, quick disconnect fasteners.

3.5.4.8.2 Class B machines.- Covers and access panels that require removal for adjustment of interior components, shall be equipped with screw-type fasteners.

* 3.6.4.9 Electrical components.- Unless otherwise specified (see 6.2), all electrical components shall operate on a 115-120 volt, 60 hertz, single-phase, alternating-current power source. The electric motors shall be continuous duty, conforming to CC-M-636 and shall be equipped with permanently lubricated sealed bearings. The motor shall be tested as specified in 4.4.6 to determine proper operation. The other motor characteristics shall be as specified (see 6.2).

3.6.4.10 Lubrication.- All moving parts requiring lubrication shall be provided with accessible lubrication fittings. Accessibility to the fittings shall be provided without the removal or adjustment of accessories parts. The housing side panels and plates may be removed to provide accessibility to lubrication areas.

3.6.4.11 Exterior dimensions.- Machines shall conform to the following maximum overall dimensions:

Height	- 25 inches
Width	- 50 inches
Depth	- 32 inches

3.6.4.12 Processing unit dimensions.- The processing unit for the type II and type III machine shall conform to the following maximum overall dimensions:

Height	- 30 inches
Width	- 44 inches
Depth	- 20 inches

3.6.4.13 Weight.- The individual weight of the type I, type II, and type III (without processing unit) machines, shall not exceed 175 pounds. The weight of the processing unit for the type III machine shall not exceed 20 pounds.

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3.7 Construction.- The class A machines shall be constructed to withstand the rigidity and transportability test specified in 4.4.2. All machine housings shall be constructed of steel specified in 3.4.2.

3.7.1 Electric.-

3.7.1.1 Wiring.- All wiring shall be color coded and all circuits shall be numbered. The class A machines shall have a laminated plastic or an etched metal wiring diagram, detailing color coded wires and identifying controls, mounted on the machine exterior above the cord outlet.

3.7.1.2 Cord.- The machines shall be equipped with a rubber covered cord, ten feet long. The cord shall be directly connected to the main "on-off" switch, and have a three-prong grounding plug.

* 3.7.1.3 Electromagnetic compatibility.- All class A machines, and when specified, (see 6.2), class B machines, shall be equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.5).

* 3.7.1.4 Fungus and moisture resistance, electrical components.- All class A machines, and when specified, (see 6.2), class B machines, shall be moisture and fungus resistance treated in accordance with MIL-T-152.

3.8 Accessory equipment.-

3.8.1 Dust cover.- A plastic dust cover shall be furnished with each machine.

3.8.2 Type lettering fonts.- When specified (see 6.2), film or disc type lettering fonts of the style, point size, and type specified shall be furnished. The fonts to be used for cartographic purposes shall include all the symbols shown on figure 1 of this specification (see 6.7).

3.9 Maintenance tools.- The supplier shall furnish a complete set of maintenance tools, as specified (see 6.2), with each machine.

3.10 Finish.- The class A machines shall have a gray enamel finish conforming to 3.4.3 applied in accordance with type A of MIL-T-704. The class B machines shall have a painted finish in accordance with the standard practice of the manufacturer.

3.11 Marking for identification.- The machines shall be marked in accordance with MIL-STD-130. The marking shall be applied to the machines on identification plates conforming to MIL-P-514.

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3.12 Instruction plates.- The machines shall be equipped with instruction plates, including warnings and cautions, describing any special or important procedures to be followed in operating and servicing the equipment. The instruction plates shall conform to MIL-P-514.

3.13 Justification charts.- Justification charts shall be supplied by the manufacturer with machines.

* 3.14 Workmanship.- The machine shall conform to the following:

(a) The finish shall be uniform and free from chips, peeling, blisters or scratches.
(b) All assemblies and components shall be free of dents and fractures, and show no signs of warping.
(c) All wires shall be unbroken, unfrayed, and completely secured to their terminals.
(d) All rough and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in accordance with table I and for defects in dimensions and weight. The class A machine shall be tested in accordance with 4.4.1 through 4.4.6 (the Government shall conduct the tests specified in 4.4.2). The class B machine shall be tested in accordance with the applicable tests specified in 4.4.3.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

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4.3.1.1 In-process inspection, class A machines.- Inspection shall be made during an appropriate phase of the manufacturing process of the class A machines to determine whether the finish and fungus and moisture resistant protection specified in 3.10 and 3.7.1.4 is completely applied. Whenever nonconformance is noted, correction shall be made to both the end item and the process.

4.3.2 Inspection of the end item.- The end item shall be inspected as specified hereinafter. The lot shall consist of all the machines of one type and class offered for inspection at one time. The sample unit shall be one completely assembled machine.

4.3.2.1 Visual examination.- The machines shall be examined in accordance with the classification of defects set forth in table I. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major, and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Visual examination

Examine	Defect	Classification Major	Minor
Finish	Type finish not as specified		X
	Color not as specified (class A machines		X
	Not uniform, chipped, peeled, blistered, or scratched		X
Type of machine	Not as specified	X	
Construction and workmanship	Any component missing, dented, or fractured	X	
Light source	Bulb broken	X	
Processing unit	Cracked or malformed	X	
Covers and access	Warped, bent, do not permit access to interior components, not secured with hand-operable quick disconnect fasteners (class A) or screws (class B)		X

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TABLE I.- Visual examination (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)			
Lubrication	Moving parts not provided with lubrication fittings	X	
	Accessibility to fittings not possible without removal or adjustment of accessories or parts	X	
Electrical	Wires frayed, broken, or not connected	X	
	Wiring not coded	X	
	Wiring diagram missing or not as specified (class A machines only)		X
	Cord not rubber covered	X	
	Cord not connected to main "on-off" switch	X	
	Plug not three prong, ground type	X	
Instruction plate	Not as specified		X
Maintenance tools	Not as specified		X
Identification marking	Missing; illegible, not as specified		X

4.3.2.2 Dimensional and weight examination.- Inspection shall be made of machines to determine compliance with dimensions specified in 3.6.4.11 and 3.6.4.12 and weights specified in 3.6.4.13. The inspection level shall be S-3 and any nonconformance to specified requirements shall be cause for rejection of the lot.

4.3.3 End item testing.- Each machine shall be tested in accordance with the applicable tests specified in 4.4.3 and 4.4.4. Any nonconformance to any specified test shall be cause for rejection of the machine. When a preproduction sample is not required, the initial production unit of the class A machines shall be tested as specified in 4.4.1 through 4.4.6 and the initial production unit of the class B machines shall be tested as specified in 4.4.3 through 4.4.6. Nonconformance shall be cause for rejection of the lot.

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4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examination	Defect
Markings exterior and interior	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners or container flaps; loose strapping; inadequate stapling; bulging or distortion of containers.

* 4.4 Tests.- The supplier shall provide all necessary facilities and supplies required for performing the tests specified herein, except the Government performed test specified in 4.4.2.

* 4.4.1 Temperature and humidity storage tests (class A only).- The machine shall be placed in a temperature test chamber at normal room temperature. The temperature in the chamber shall be raised to 155 F. and maintained at this temperature for not less than 4 hours. The temperature shall then be allowed to return to normal room temperature and the machine examined for damage. The temperature in the chamber shall then be lowered to minus 65 F. and maintained for not less than 72 hours. The chamber shall then be allowed to return to normal temperature and the machine re-examined for damage. The machine shall be subjected to a high humidity test of 98 percent at 80 F. for 8 hours a day for 3 days. The machine shall be operated and examined after each 8 hour test to determine compliance with 3.4.1 and 3.5.1. Any nonconformance shall constitute failure of this test.

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* 4.4.2 Rigidity and transportability test (class A only).- The machine shall be mounted in a truck and the truck transported over a test course for not less than 20 miles. The vehicle shall be driven at top legal speed for this type equipment. Not less than 50 percent of the test course shall be unimproved roads such as earth or partially surfaced gravel, and the remainder shall be improved roads such as concrete, asphalt, or macadam. The test shall include not less than 20 quick starts and stops of the vehicle on grades of not less than 5 percent or greater than 10 percent. Fifty percent of the starts and stops shall be made on ascending grades and fifty percent on descending grades. Upon completion of this test, the machine shall be subjected to the performance tests in 4.4.3. Any nonconformance shall constitute failure of this test.

4.4.3 Performance tests.-

* 4.4.3.1 Performance with photosensitized paper.- The machines shall be set-up using photosensitized paper, in a fully lighted room at 72 F. and 45 percent relative humidity, to test for photographic and mechanical performance. There shall be three phases of this test: the first, using a 10 point type lettering font; the second, using a 72 point lettering font; and the third, using a 24 point, Thai foreign language type lettering font (see 6.6). The fonts, except for the Thai font, shall be English or similar, and shall contain upper and lower case letters. During each phase of the test, all controls and adjustments shall be operated. Fifty characters shall be printed with the single line and multi-line machines for each phase. In addition, fifteen characters each shall be printed on three lines with the multi-line machines using a 10 point type lettering font. The paper and composition shall be examined at the completion of each phase to determine compliance with the following paragraphs of this specification, as applicable: 3.5, 3.6.1, 3.6.2, 3.6.3, 3.6.4.1, 3.6.4.2, 3.6.4.2.1, 3.6.4.3, 3.6.4.4, 3.6.4.5.1, 3.6.4.6.1, and 3.6.4.6.2. Any nonconformance shall constitute failure of this test.

* 4.4.3.2 Performance with film.- The machine shall be set- p as specified in 4.4.3.1 to determine compliance with the same applicable paragraphs, except film shall be used. There shall be three phases of this test also, using the same procedures as the test specified in 4.4.3.1. When performing this test on type I and type II machines, a measurement of the amount of exposed film (as a result of installing the roll) shall be made to determine compliance with 3.6.4.5.1. Any nonconformance shall constitute failure of this test.

4.4.3.3 Processing unit (type III machine).- The processing unit for the type III machine shall be fed 2 feet of exposed, 8 inch wide, photosensitized paper in a lighted room at ambient 70 F. to 80 F., and 40 to 50 percent

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relative humidity. The time required for the developed copy to appear shall be observed to determine compliance with 3.6.4.6.3. Any nonconformance shall constitute failure of this test.

* 4.4.4 Exposure meter test.- The exposure meter shall be tested by energizing the machine and observing the exposure meter reading. The adjusting rheostat shall then be turned both clockwise and counterclockwise to determine compliance with 3.6.4.2.2. Any nonconformance shall constitute failure of this test.

* 4.4.5 Electromagnetic compatibility.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report, and the requirements meet MIL-STD-461.

* 4.4.6 Operational test.- The machine shall be connected to an alternating current source compatible with the electrical characteristics of the machine being tested. The "on-off" switch shall be turned to the "on" position. All the controls shall be operated and checked for freedom of movement to determine compliance with 3.5. The motor shall be operated to determine compliance with 3.6.4.9. Any nonconformance shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Composing machines shall be preserved and packaged method II, and manuals packaged method IC in accordance with MIL-P-116.

* 5.1.2 Level C.- Composing machines shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Each composing machine, together with equipment manual and accessory equipment, preserved and packaged as specified in 5.1, shall be packed in a fiberboard, nailed wood, or cleated plywood shipping container

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conforming to grade V2s, style RSC of PPP-B-636; class 2, style 2 or 4 of PPP-B-621; or overseas type of PPP-B-601. The weight and dimensional limitations of the applicable container specification shall not be exceeded. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.2.2 Level B.- Each composing machine, together with equipment manual and accessory equipment, preserved and packaged as specified in 5.1, shall be packed in a fiberboard, nailed wood, cleated plywood or triple-wall fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-621; domestic type, style A or B of PPP-B-601; or class 1, grade A, style E of PPP-B-640, respectively. The weight and dimensional limitations of the applicable container specification shall not be exceeded. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

* 5.2.3 Level C.- Composing machines, together with equipment manuals and accessory equipment, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in general purpose photo composing operations. The class A machine is intended for field use in the preparation of propaganda leaflets and cartographic typesetting operations. The class B machine is intended for office use in the preparation of display and headline type.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of machines required (see 1.2).
(c) When a first article is not required (see 3.1, 4.2, and 6.3).
(d) Whether the machine is for 35 or 51 millimeter paper and film (see 3.6.1 and 3.6.2).

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(e) Whether table or floor model type II machine is required (see 3.6.2).
(f) Whether integral or external type processing unit is required (see 3.6.2).
(g) When an exposure meter is required (see 3.6.4.2.2).
(h) Whether automatic or manual film and paper processing is required (see 3.6.4.6.2).
(i) Electrical characteristics, if other than specified (see 3.6.4.9).
(j) Motor characteristics required (see 3.6.4.9).
(k) When electromagnetic compatibility is required for class B machines (see 3.7.1.3).
(l) When class B machines shall be fungus and moisture resistant treated (see 3.7.1.4).
(m) Style, point size, and type (film or disc) of lettering font, when font is required (see 3.8.2).
(n) Maintenance tools required (see 3.9).
(o) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

* 6.2.1 Contract data requirements.- Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

* 6.3 First article.- The item covered by this specification requires first article inspection and approval under the appropriate provision of Paragraph 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.4. Printing supplies.- All necessary printing supplies shall be obtained separately since they are not included as components of this specification.

* 6.5 Government performed tests.- The supplier shall arrange with the contracting officer for rigidity and transportability tests of the preproduction sample or initial production unit, as applicable (see 4.4.2).

* 6.6 Foreign language font.- A Thai foreign language lettering 24 point font may be obtained from the Director, General Equipment and Packaging Laboratory, U.S. Army Natick Laboratories, Kansas Street, Natick, MA 01760.

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* 6.7 Copies of Army Map Service (AMS) patterns and AMS cut numbers may be obtained from the Corps of Engineers, U. S. Army Map Service, Washington, D. C. 20314.

6.8 Photosensitized paper and film, 35 mm.- Thirty-five (35) mm photosensitized paper and film are available with a plain back or adhesive back.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

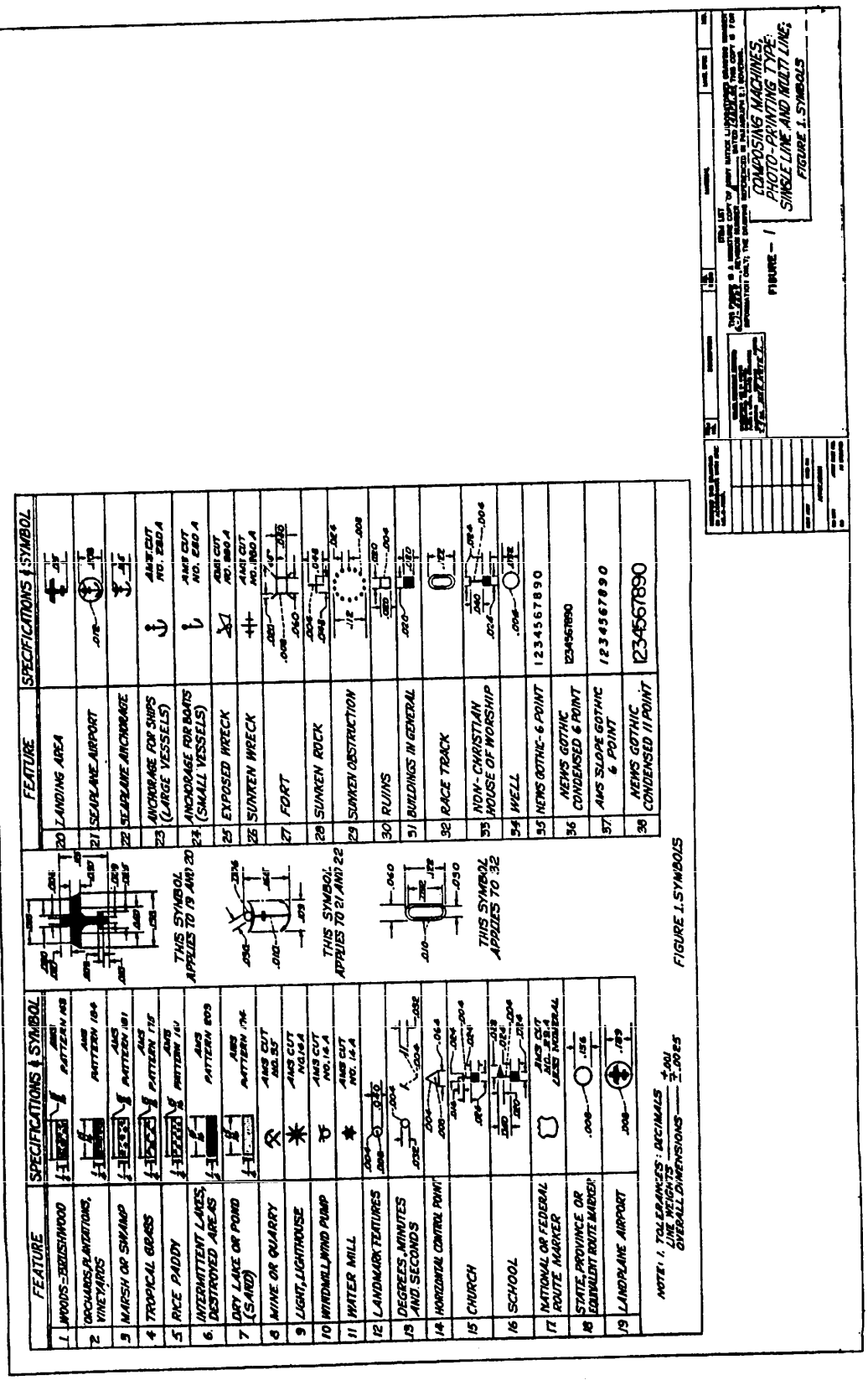
Army - GL

Project No. 7430-0019

Review activity:

Army - ME

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Composing Machines, Photo-Printing Type: Single Line And Multi-Line MIL-C-43232A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

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